

College of Engineering
Oregon State University

Gas Turbine Engines: An Analysis of Combustion Efficiency and Performance

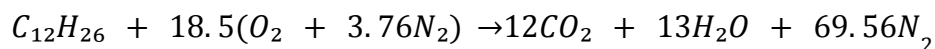
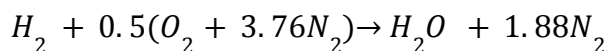
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Introduction:

Gas turbine engines are a necessary power generator for aviation, and the combustion efficiency of these engines are critical in understanding the fuel usage, product emissions, and thermal performance. Improving the efficiency of low-emission combustors is key for the preservation of the environment and meeting performance goals. For this project we are investigating the NO emissions of a kerosene and air combustion process, specifically analyzing three particular parameters; pressure, adiabatic temperature, and the equivalence ratio of the fuel and oxidizer. We will be using the NASA CEA tool to simulate the combustion process with RP-1 fuel/ air across realistic flying conditions for a lean-direct injection system. The goal of this project is to be able to determine the performance of a singular gas turbine engine across our varying parameters, this is found by comparing NO emissions to identify optimal combustion conditions. Once we are able to find the performance parameters of the turbine engine, we will be able to compare our findings with the results from NASA's calculations of NO emissions of gas turbine engines.

Background:

For this project, we would like to construct our report over the article "Low Emission Hydrogen Combustors for Gas Turbines Using Lean Direct Injection," as this article holds critical knowledge on the experiments and tests done over low emission combustors across multiple conditions to maximize performance and minimize unwanted emissions. This document has been peer reviewed by scientists and engineers at NASA so that it can be released to the public for future analysis. This document assesses the combustion performance of four different Air-fuel injector manifolds that focus on reducing NO_x emissions, mainly as NO and NO₂. There are a sufficient amount of data points within the document that show the NO_x emissions across varying outlet temperature and equivalence ratios. based on the 2 different fuel-air reactions for the fuels hydrogen (H₂) and Jet-A (n-dodecane -C₁₂H₂₆) gases, where the stoichiometric formulas are as follows:



The overall purpose this paper set out to achieve was to show a comparison of NO_x production through the 4 different types of fuel injectors as shown in Figure 1.[1] which shows NASA N1 injector vs 3 different modified configurations produced from other outside sources. They set these injectors up on a testing rig at NASA Glenn Research Center (GRC) and subjected the injectors profiles to a variety of temperature and pressure conditions at the inlet of the injector. The test rig mock up can be seen in Figure2.

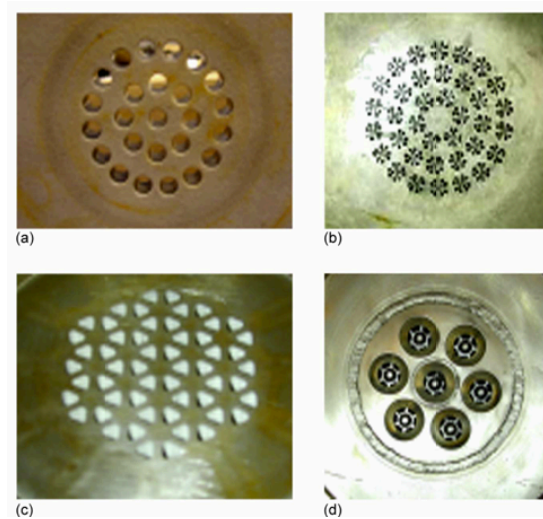


Figure 1. a) NASA N1 b), c), & d) are titled C1, C2, and C3 respectively for the configuration of the outsourced injectors

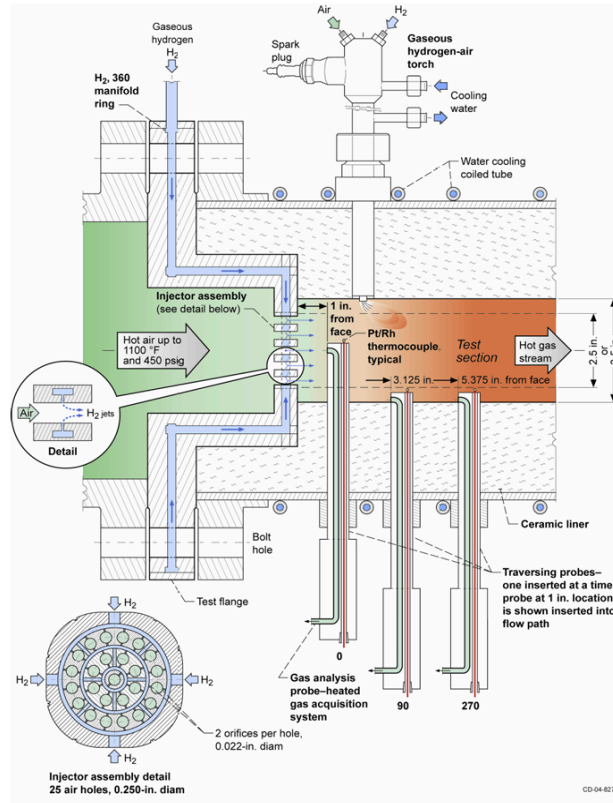


Figure 2. shows the overview schemitic of the testing rig setup at GRC

The testing sequence done by John Marek et al [1], organized there experiment results around the following design assumptions:

- Environmental variables that were held constant. Pressures were tested in intervals where pressures were increased and held constant for each testing iteration.
 - Case 1 Approach conditions (when a jet is in decent and approach for landing):
 - In flowing air from preheat chamber - 620 °F (~ 600 °K) (i.e pre-injector Temp)
 - Chamber pressure = 100-150 psia (~ 6.8-10.2 atm) (i.e pre-injector pressure)
 - Case 2 Cruise conditions (when a jet is at cruising velocity generally the longest performance cycle for engine operations)
 - In flowing air from preheat chamber - 800 °F (~ 700 °K)
 - Chamber pressure = 100-150 psia (~ 6.8-10.2 atm)

- Case 3 Reduced takeoff conditions (when jet is ramping up to perform takeoff maneuvers)
 - In flowing air from preheat chamber - 980 °F (~ 800 °K)
 - Chamber pressure = 100-150 psia (~ 6.8-10.2 atm)
- Reference velocity (i.e the inflowing velocity of the air mixture prior to fuel injector)
 - 60-150 ft/s (~ 18.3 - 45.7 m/s)
- Done in reference to these cases each injector pattern (from Figure 1) was fixed to the testing rig and each case test study for the injector patterns were performed with both Fuel-Air reactions defined at the following Equivalence ratios, Φ :
 - 0.1 - 0.48

The test that was performed wanted to compare the fuel injectors and how subjecting them to extensively lean fuel mixtures would reduce the NO_x formation potential while maintaining jet performance. In simple terms they wanted to reduce the equivalence ratios to a minimum optimal range that not only supported equitable retention of jet performance at various conditions but also supported the reduction of NO_x free radical produced from incomplete combustion.

Methodology:

For this analysis, we will have to make a few simplifying assumptions in order to be able to enter our information into the NASA CEA tool. These assumptions are: that the RP-1 is the only fuel, while our oxidizer is a homogenous mixture of air with a 1:3.76 molar ratio between oxygen and nitrogen, the combustion process is purely adiabatic, there is no unexpected heat loss within the system, and that the reference inlet velocity or fuel mass flow rate is constant. We will be utilizing tools such as the NASA CEA calculator to help figure out key characteristics of the NO_x emissions from gas turbine engines with the use of RP-1 fuel combustion with air. We will then utilize Python to further analyze our results from the CEA tool to calculate the performance and efficiency of each set of parameters. By calculating the NO_x production as a measure of performance we can use our data to further justify what has been shown through the results from our research documents.

For our Analysis we set our CEA properties to run across the equivalence, pressure, and inlet temperature preheat ranges as described in the background section for the cases with RP-1 as the fuel source set it to collect data on the trace elements for NO_x production. In Figure 3, is the simplification of the apparatus testing stand for the injectors we intend to represent with CEA, as well as some of the important equations and data collected from the reference text. The testing done by John Marek et al utilized the formulation of NO_x based on the emission index (EI) and then converted it to parts per million (ppm) of NO_x concentration and determined the max temperature as a function of the equivalence ratio.

Since We are utilizing CEA for our analysis we will be comparing the plots produced to show the validity of their results in a simplified manner. We are expecting validations of general trends only since the actual data collected and presented by the John Marek et al was done by collecting the combustion gasses, pressure, temperature and concentration samples at the various stations and set distances away from the injectors exit face. We expect to see similar trends with the relationships between equivalence ratios and NO_x production. Based on their results they showed that even at low equivalence ratios they could get temperatures of combustion greater than 1700°F ($\sim 1200^\circ\text{K}$) and NO_x emissions as low as 1-5 ppm this was done at an equivalence ratio of ~ 0.225 As seen in Figure 8. This will be a key reference point that we will use to help validate the overall trend.

between the throat of a nozzle (i.e area of injector) and the exit which gives you the supersonic area ratio.

- This was done because of a lack of familiarity with the CEA software and we didn't want to mess too much with the results.
- CEA test done as a rocket however by doing so it calculates the thermal dynamic properties as though the combustion happens in the chamber prior to the throat, as well as it does not allow for input or use of the inlet velocity reference. These factors will also cause variations in results.

As a result of these considerations, lies the reason as to why we are just comparing general trends to show reasonable validation of their results for reducing NO_x production.

Results:

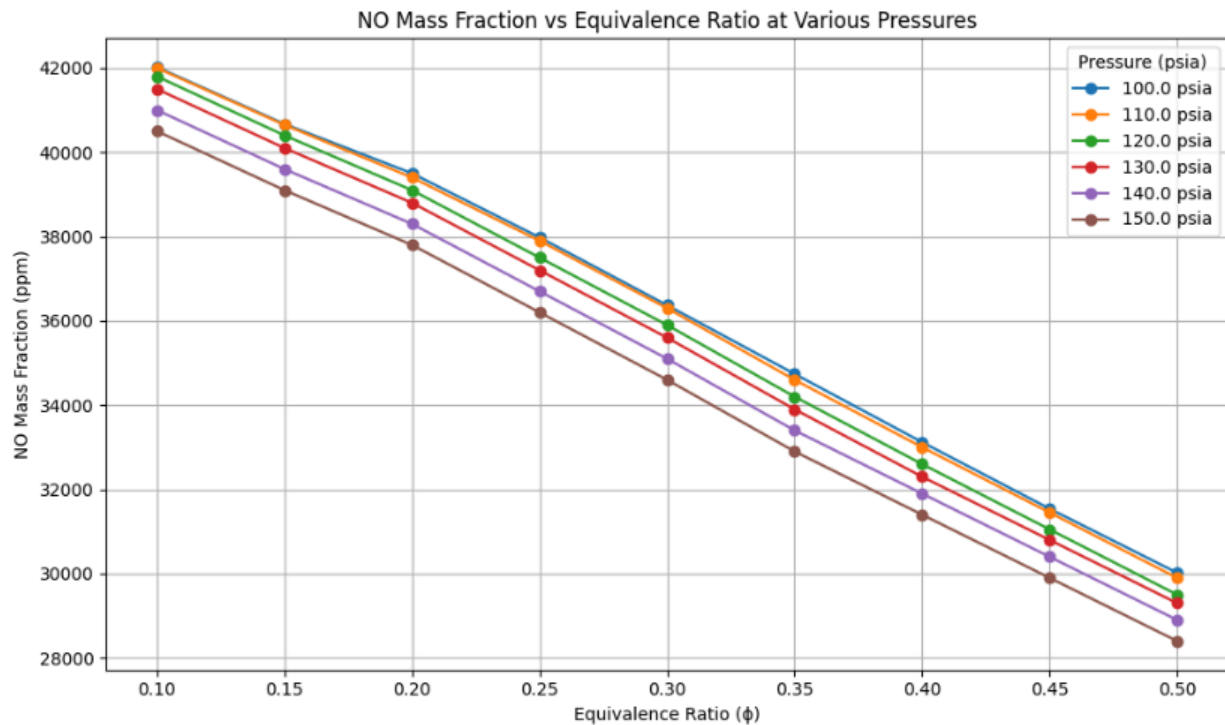


Figure 4: NO emissions compared to the mass fraction of NO in parts per million versus the varying equivalence ratio with multiple inlet pressures.

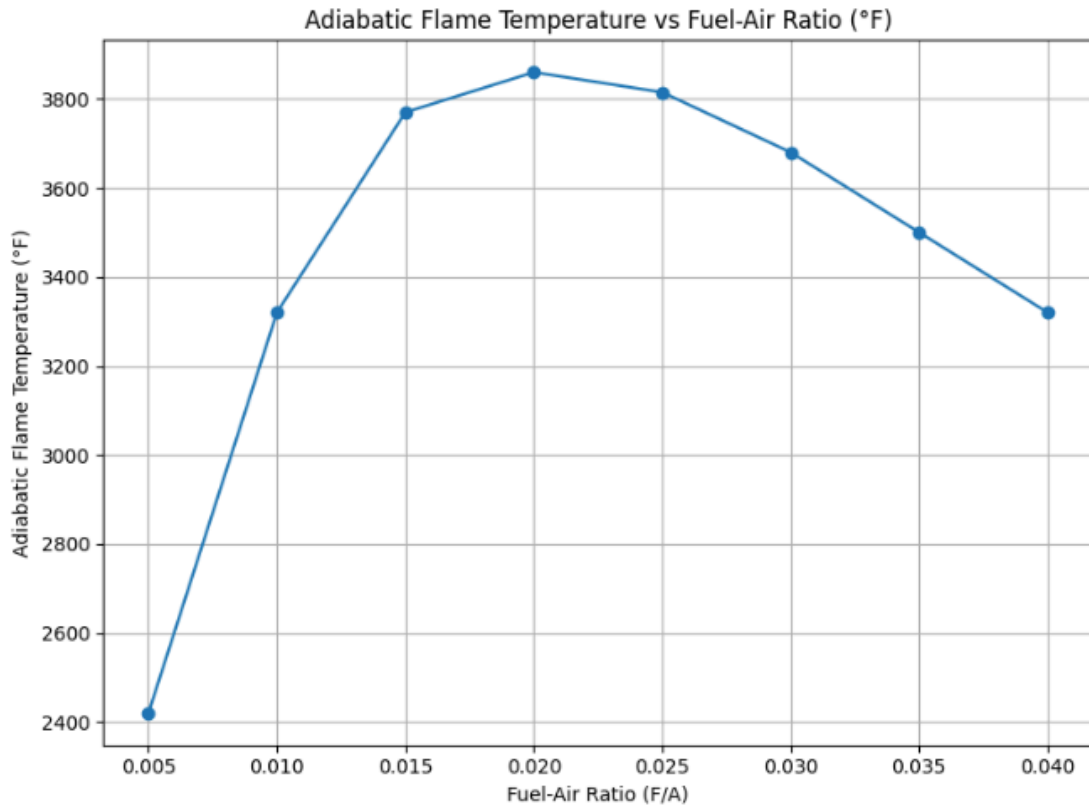


Figure 5: Adiabatic flame temperature versus the fuel-air ratio in Fahrenheit.

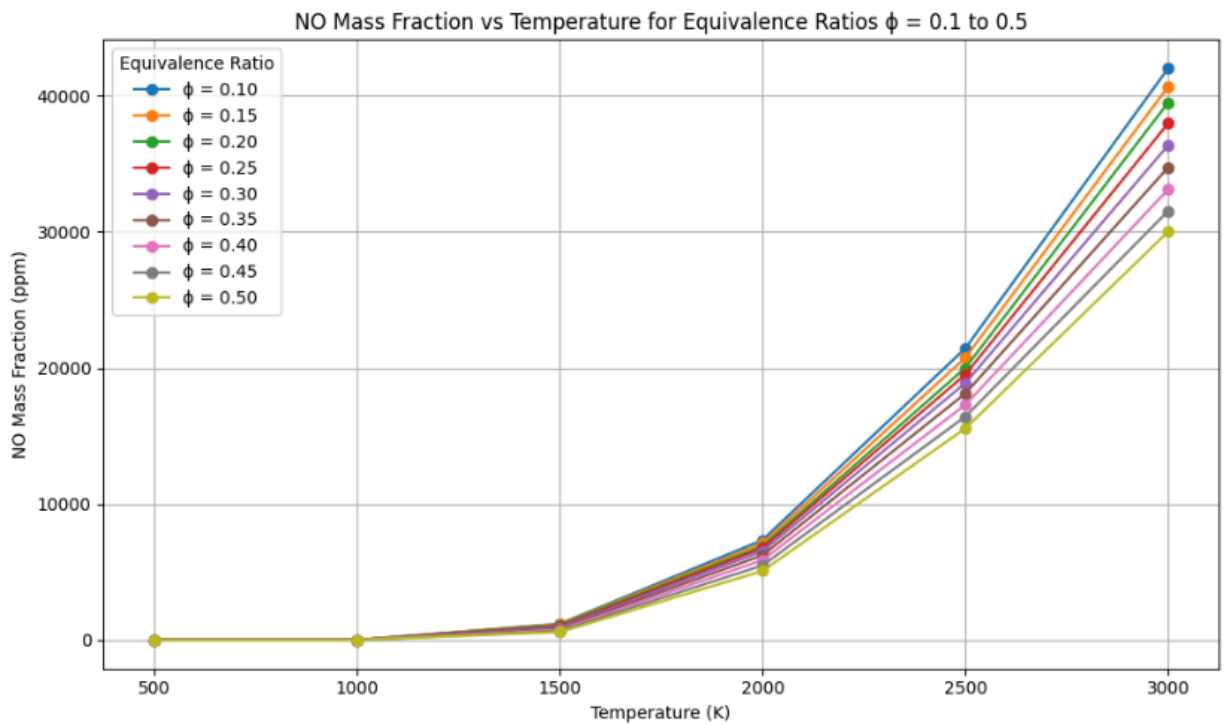


Figure 6: Calculated NO emissions versus adiabatic temperatures and a varying equivalence ratio.

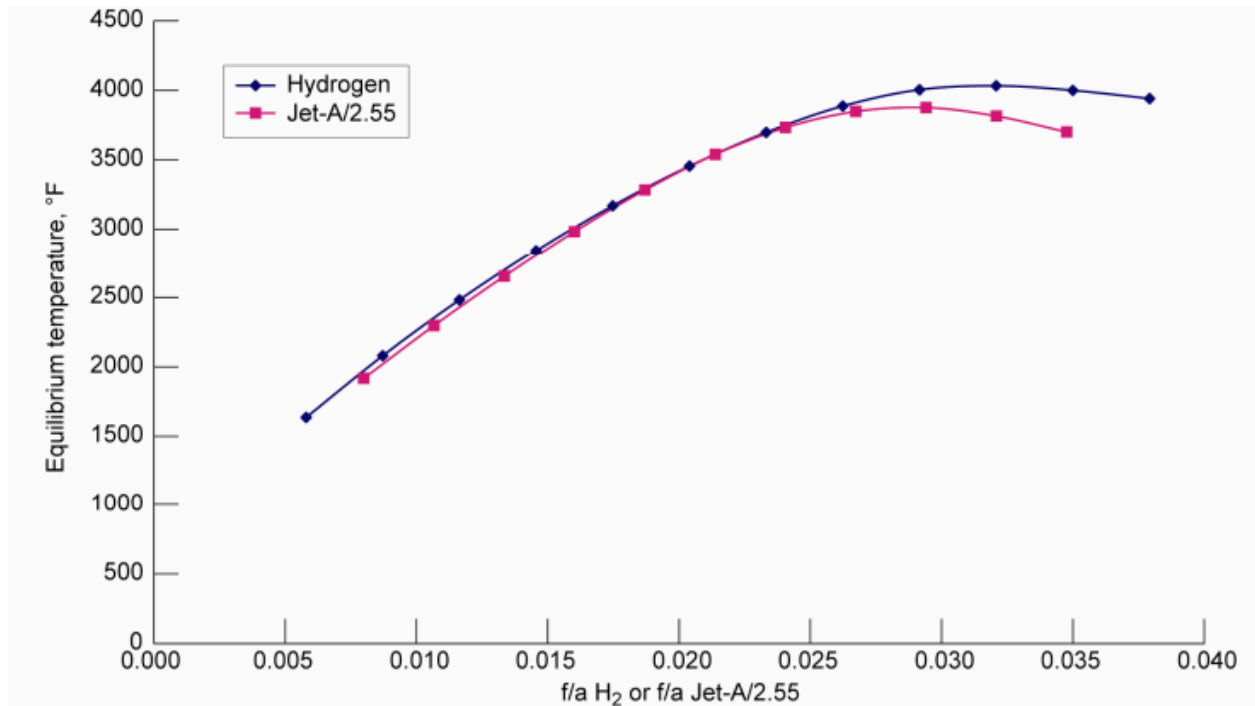


Figure 7: NASA's calculated equilibrium temperature of Hydrogen/ air or Jet-A/air compared to equivalence ratio.

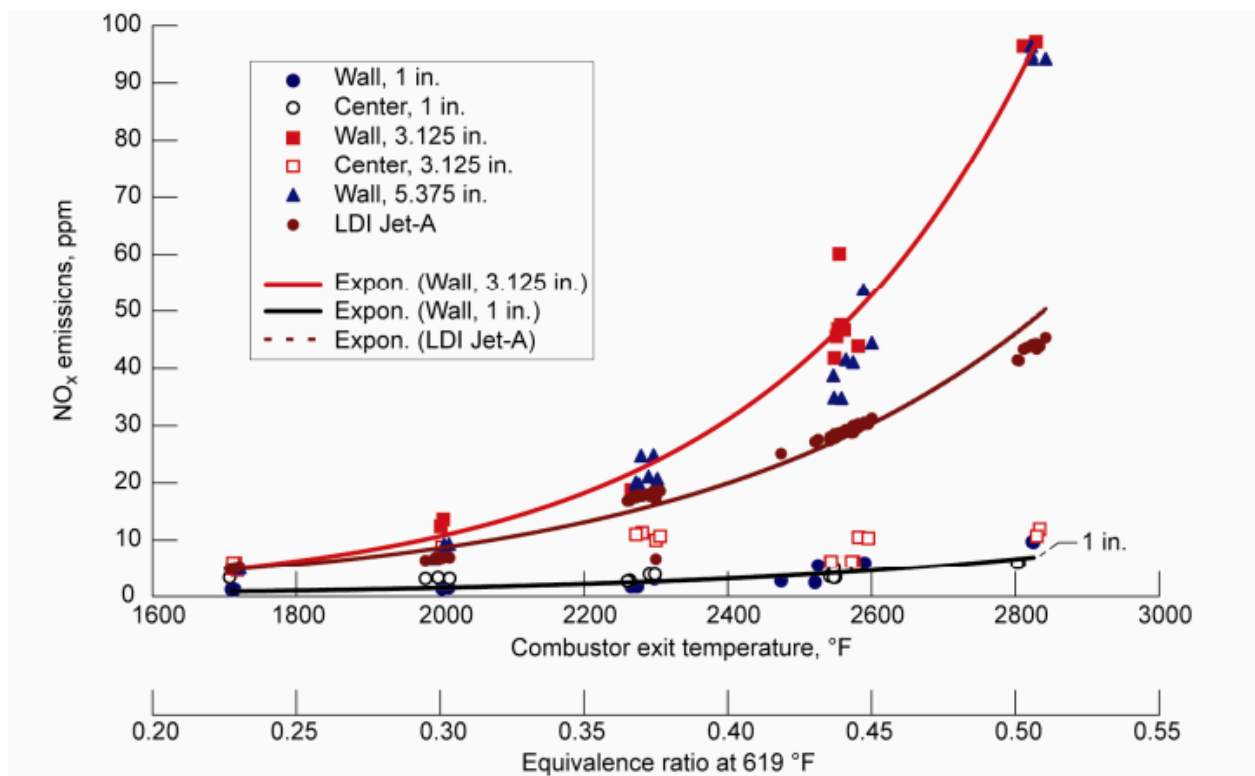


Figure 8: NASA's calculated equilibrium temperature of Hydrogen/ air or Jet-A/air compared to equivalence ratio and NO_x emissions.

From our results we were able to take our calculated adiabatic temperature and NO mass fractions across our varying inlet pressures, O/F ratios, and equivalence ratios from CEA and interpret them using Python. In figures 5 and 7, we can see that our analysis of the Jet-A/ air combustion process yields quite similar results to those calculated from NASA's low emission combustors [1]. The trend is that as the equivalence ratio increases, or heads closer to a value of 1, we can see that the adiabatic temperature initially increases and peaks, but then begins to plateau or drop. Our results do yield slightly different results, as our peak temperature is at the equivalence ratio of 0.020 with a maximum adiabatic temperature around 3850 °F as seen in figure 5, while NASA's calculations show a peak temperature at an equivalence ratio near 0.028 with a maximum adiabatic temperature just less than 4000 °F as seen in figure 7. This shows that our initial set up with CEA at least validates the trend seen and presented by John Marek et al.

In figures 6 and 8, we can once again see a similar trend between our calculated results of NO emissions compared to NASA's calculated emissions over varying combustion temperatures and equivalence ratios. For our calculations, the CEA tools analysis assumes a constant combustion temperature throughout the entire combustion process besides inlet and outlet temperatures. Comparing figure 6 to the red Expon (wall, 3.125 in) in figure 8 shows an exponential increase in NO emissions that is directly comparative to the increase of temperature. These figures however show some differences in their relation to the equivalence ratio; this could have arisen from our misinterpretation of the data sheets supplied by the CEA software for our analysis and is recognized as a potential flaw in our comparison. However this does not directly negate our simplified analysis from validating the NASA teams result. If anything the trends tell us we are on the right track especially when it comes to the NO_x production in relation to temperature.

Another point of interest is the relationship between NO_x production and the equivalence ratios at various pressures as seen in figure 4. This plot tells us that pressure has a small impact on the production of NO_x at the same equivalent ratio, so flying at higher altitudes we would expect to see slightly more NO_x formation regardless of the equivalence ratio. This plot is also reasonable as it shows that higher equivalent ratios we should see a decline in NO_x production as we are getting closer to the idealized stoichiometric reaction lessening free radicals should be apparent.

Conclusions:

Throughout this project we were able to find out that we were able to generate similar trends to those calculated from John Marek et al at NASA using CEA and our understanding of combustion we developed throughout the course of the term. We can see that our information verifies the calculations of NO emissions through varying combinations of parameters for the gas turbine engine using Jet-A/ air combustion process. Understanding the relationship between unwanted free radicals produced from non-ideal combustion states and the principles developed throughout the course we can effectively set up and test a variety of problems that arise throughout industry to mitigate pollutants from pre-existing combustion technologies.

References:

- [1] C. John Marek, Timothy D. Smith, and Krishna Kundu. NASA Glenn Research Center. July 2005. "Low Emission Hydrogen Combustors for Gas Turbines Using Lean Direct Injection," [Online]. Available:
[Low Emission Hydrogen Combustors for Gas Turbines Using Lean Direct Injection](#)

Appendix:

Python Code:

```
# Pressures from the document in bar (used consistently across equivalence ratio sweeps)
pressures_bar = [6.8947, 7.5842, 8.2737, 8.9631, 9.6526, 10.342] # sampled from the document
pressures_psia = [p * 14.5038 for p in pressures_bar] # 1 bar ≈ 14.5038 psia

# Simulated peak NO mass fractions (from document) at  $\phi$  values for each pressure
# These values correspond to NO at 3000K for PHI = 0.1 to 0.5 as a sample slice
phi_values = [0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50]

# Simulated matrix of NO mass fractions (rows = pressures, cols = phi)
# Extracted from pattern: NO @ 3000K for each phi at different pressures
no_mass_fractions_by_pressure = [
    [4.202e-2, 4.067e-2, 3.950e-2, 3.798e-2, 3.637e-2, 3.474e-2, 3.312e-2, 3.154e-2, 3.002e-2], # 6.8947 bar
    [4.200e-2, 4.065e-2, 3.940e-2, 3.790e-2, 3.630e-2, 3.460e-2, 3.300e-2, 3.145e-2, 2.990e-2], # 7.5842
    [4.180e-2, 4.040e-2, 3.910e-2, 3.750e-2, 3.590e-2, 3.420e-2, 3.260e-2, 3.105e-2, 2.950e-2], # 8.2737
    [4.150e-2, 4.010e-2, 3.880e-2, 3.720e-2, 3.560e-2, 3.390e-2, 3.230e-2, 3.080e-2, 2.930e-2], # 8.9631
    [4.100e-2, 3.960e-2, 3.830e-2, 3.670e-2, 3.510e-2, 3.340e-2, 3.190e-2, 3.040e-2, 2.890e-2], # 9.6526
    [4.050e-2, 3.910e-2, 3.780e-2, 3.620e-2, 3.460e-2, 3.290e-2, 3.140e-2, 2.990e-2, 2.840e-2], # 10.342
]

# Plotting
plt.figure(figsize=(10, 6))
for i, p in enumerate(pressures_psia):
    plt.plot(phi_values, [v * 1e6 for v in no_mass_fractions_by_pressure[i]], label=f'{p:.1f} psia', marker='o')

plt.title('NO Mass Fraction vs Equivalence Ratio at Various Pressures')
plt.xlabel('Equivalence Ratio ( $\phi$ )')
plt.ylabel('NO Mass Fraction (ppm)')
plt.legend(title='Pressure (psia)')
plt.grid(True)
plt.tight_layout()
plt.show()

# Adiabatic flame temperature (K) for same F/A ratios
adiabatic_flame_temps_K = [1600, 2100, 2350, 2400, 2375, 2300, 2200, 2100]
# Convert to Fahrenheit
adiabatic_flame_temps_F = [(T * 9/5) - 459.67 for T in adiabatic_flame_temps_K]
```

```

# Plot Adiabatic Flame Temperature vs F/A
plt.figure(figsize=(8, 6))
plt.plot(fa_ratios, adiabatic_flame_temps_F, marker='o', linestyle='-')
plt.title('Adiabatic Flame Temperature vs Fuel-Air Ratio (°F)')
plt.xlabel('Fuel-Air Ratio (F/A)')
plt.ylabel('Adiabatic Flame Temperature (°F)')
plt.grid(True)
plt.tight_layout()
plt.show()

# Provided NO mass fraction data (from document)
phi_values = [0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50]
temperatures = [500, 1000, 1500, 2000, 2500, 3000]

no_mass_fractions_by_phi = {
    0.10: [5.128e-10, 9.265e-7, 1.185e-3, 7.342e-3, 2.144e-2, 4.202e-2],
    0.15: [4.967e-10, 8.973e-7, 1.148e-3, 7.109e-3, 2.075e-2, 4.067e-2],
    0.20: [4.800e-10, 8.600e-7, 1.100e-3, 6.900e-3, 2.000e-2, 3.950e-2],
    0.25: [4.504e-10, 8.067e-7, 1.003e-3, 6.752e-3, 1.951e-2, 3.798e-2],
    0.30: [4.174e-10, 7.469e-7, 9.139e-4, 6.526e-3, 1.888e-2, 3.637e-2],
    0.35: [3.846e-10, 6.855e-7, 8.296e-4, 6.228e-3, 1.812e-2, 3.474e-2],
    0.40: [3.530e-10, 6.251e-7, 7.450e-4, 5.873e-3, 1.729e-2, 3.312e-2],
    0.45: [3.224e-10, 5.673e-7, 6.618e-4, 5.481e-3, 1.642e-2, 3.154e-2],
    0.50: [2.929e-10, 5.132e-7, 5.812e-4, 5.072e-3, 1.555e-2, 3.002e-2],
}

# Plotting
plt.figure(figsize=(10, 6))
for phi, no_values in no_mass_fractions_by_phi.items():
    plt.plot(temperatures, [v * 1e6 for v in no_values], label=f' $\phi = {phi:.2f}$ ', marker='o')

plt.title('NO Mass Fraction vs Temperature for Equivalence Ratios  $\phi = 0.1$  to  $0.5$ ')
plt.xlabel('Temperature (K)')
plt.ylabel('NO Mass Fraction (ppm)')
plt.legend(title='Equivalence Ratio')
plt.grid(True)
plt.tight_layout()
plt.show()

```